

Numerical analysis of quantum many-body scars and many-body localized discrete time crystals

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(Dated: 2026-10-01)

I. INTRODUCTION

Both many-body localized discrete time crystal (MBL-DTC), and quantum many body scars (QMBS), involve revivals of an initial state during time evolution. In the context of MBL-DTC, this is interpreted as period doubling or symmetry breaking of time-translational symmetry as the initial state is revived after two periods of the drive. In the context of QMBS, this is interpreted as the result of constrained dynamics due to the nearest neighbor Rydberg blockade condition. In both cases, there is a non-negligible $\gg \mathcal{O}(2^{-n})$, probability of returning to the initial state.

In MBL-DTC, the revivals are attributed to eigenstate order while in QMBS, they are attributed to an approximate block diagonal structure of the Hamiltonian. Perhaps, the approximate block diagonal structure leads to revival probabilities being close to but not equal to 1.

In QMBS, there is a special band of energy eigenstates in the so-called scarred subspace, which has properties that are distinct from the rest of the energy eigenstates. These special band eigenstates have energy, not just at the edges, but throughout the spectrum. These eigenstates have surprisingly low entanglement entropies, as would *not* be expected from the properties of a chaotic Hamiltonian.

In MBL-DTC, the dynamics are determined by the Floquet operator of one period of the drive. The eigenvalues of the Floquet operator are unimodular complex numbers, and in the period-doubling MBL-DTC phase they occur in pairs differing by a phase of π . This pairing is the eigenstate-order signature of the phase [1, 2]: it is exact in the ideal-pulse soluble limit, and emergent though rigid for generic disorder [3], rather than a symmetry identity of the driven Hamiltonian. Unlike QMBS, there is no special band and the eigenstate order exists throughout the spectrum.

This manuscript describes the models and the numerical methods, implemented in the `mbL.eigen` package accompanying this manuscript,[4] used to study the eigenvalue structure and eigenstate ordering associated with these two settings, together with the results obtained so far and the analyses that remain to be completed.

II. REVIEW

The QMBS phenomenology originates in the observation of long-lived revivals of the Néel ordered initial state in the 51-atom Rydberg quantum simulator of Ref. [5], and the controlled driven dynamics of Ref. [6] subsequently established tunable access to this regime. The PXP Hamiltonian that describes the blockade-constrained dynamics was then shown to exhibit a weak breaking of ergodicity: a tower of special eigenstates, not concentrated at the spectral edges but spread through the spectrum, carries anomalously low entanglement entropies and distinct spectral statistics [7, 8]. The scarred states are nearly equally spaced in energy, and have been reinterpreted as finite-energy-density condensates of weakly interacting π -magnons [9]. Exact scarred eigenstates of the PXP model were constructed in Ref. [10], the scarred dynamics were shown to be governed by an emergent $SU(2)$ symmetry with fine-tuned deformations producing perfect scars [11], related constructions cover spin-1 XY magnets with subextensive entanglement and spatiotemporal long-range order [12] and Rydberg models in the antiblockade regime with emergent kinetic constraints and finite-entanglement revivals [13], and scar states were identified in the deconfined regime of lattice gauge theories, where extensive local constraints play the role of the blockade condition [14]. The weak-ergodicity-breaking picture of Ref. [15] and the quasiparticle perspective of Ref. [16] unify these developments, treating scars as stable quasiparticle species throughout the energy spectrum and placing them alongside symmetry-based constructions, projector embeddings, and Hilbert-space fragmentation.

These eigenstates are significant as violations of expectations encoded in the eigenstate thermalization hypothesis, under which the eigenstates of a chaotic Hamiltonian behave as random variables, so that an initial state built from them approaches thermal equilibrium [17]. A quantitative benchmark is given by the Page value: the subsystem entropy of a typical pure state of a bipartite system is close to that of the maximally mixed reduced state [18]. The entanglement entropy of eigenstates and the mean adjacent level-spacing ratio, the two diagnostics computed in the workflows of Secs. V A and V C, are the standard tools for locating deviations from these expectations. The level-spacing ratio is insensitive to the local density of states and therefore applies to many-body spectra without unfolding [19]; the crossover it detects, from Wigner-Dyson statistics characteristic of chaotic spectra to Poisson statistics of integrable or localized ones, has been used to diagnose many-body localization since Ref. [20].

The MBL-DTC descends from the Floquet phases with spatiotemporal long-range order identified in Refs. [1, 2]: in a periodically driven, strongly disordered spin system, the Floquet eigenstates occur in cat doublets separated by a quasienergy difference of π , and this eigenstate order underlies the period-doubled response of local observables [1]. The pairing is stable to all weak local deformations of the drive, including ones that explicitly break the defining symmetries of the soluble pulse models, which occupy submanifolds of a larger absolutely stable phase [21]; this rigidity, together with the criticality and candidate realizations of the phase, was analyzed in Ref. [3]. The stabilizing mechanism is many-body localization, whose phenomenology, including quasi-local integrals of motion and the spreading of entanglement after quenches, is reviewed in Ref. [22]. Experimentally, discrete time-crystalline order was observed in a disordered dipolar many-body system [23], in a trapped-ion spin chain [24], in an ordered dipolar system [25], in a programmable spin-based quantum simulator [26], and on a superconducting quantum processor [27]; a disorder-free prethermal variant, stabilized by high-frequency driving rather than by localization, was observed in a trapped-ion simulator [28].

Interpreting propagator eigenvalues requires distinguishing two pairing mechanisms, which I analyze in Secs. III and IV. A chiral, or particle-hole, symmetry of the Hamiltonian forces the spectrum to be symmetric under $E \leftrightarrow -E$, so that propagator eigenvalues occur in complex-conjugate pairs $\eta \leftrightarrow \eta^*$; at zero detuning the PXP model realizes this mechanism through the anticommuting particle-hole operator $\prod_l Z_l$ [7], and for Floquet unitaries chiral symmetry likewise produces exact scarred multiplets [29]. The period-doubled MBL-DTC instead pairs quasienergies as $\varepsilon \leftrightarrow \varepsilon + \pi$, so that propagator eigenvalues occur in phase- π pairs $\eta \leftrightarrow -\eta$; this cat-doublet structure is an emergent eigenstate-order signature rather than a symmetry identity of the driven Hamiltonian [1–3]. The two pairings coincide only at special points, such as the DTC point of the dipolar experiment of Ref. [23], where the paired quasienergies cluster near $\pi/2$; away from such points the distinction is sharp, and conflating the two would misattribute a symmetry consequence to eigenstate order, or conversely.

The numerical methodology of this work draws on established traditions. The level-spacing statistics of Sec. V C and the half-chain entanglement entropies compared against the maximally mixed reference value in Sec. V A follow the diagnostic program outlined above [18–20]. The circuit layer of Sec. V D continues a line of work in which scar and time-crystal dynamics are implemented on gate-based quantum computers: time-crystalline eigenstate order was probed on a superconducting quantum processor [27], and Trotterized simulations of scar dynamics in Rydberg-blockade (PXP) chains have been demonstrated on quantum hardware with pulse-level control and error mitigation [30].

Relative to this literature, the contribution of the present manuscript is deliberately methodological. I do not propose new models and I do not claim new eigenstate-order physics; rather, the `mbL_eigen` package implements, in one reusable code base, the model builders of Secs. III and IV, eigensolver backends spanning CPU and accelerator devices, level-spacing statistics for both energy and eigenphase spectra, entanglement diagnostics, and a Qiskit circuit and simulation layer for the same propagators. The Floquet operator of the MBL-DTC is constructed gate-natively, without Trotterization, while the random-field MBL dynamics is synthesized as a first- or second-order Trotter circuit, and the output conventions of Sec. V E, including seeded disorder sampling and regression tests, make the figures of this manuscript traceable to the code that produced them. The remainder of the manuscript documents these methods in detail and the results obtained with them so far.

III. MANY-BODY LOCALIZED DISCRETE TIME CRYSTAL

Many-body localized discrete time crystal (MBL-DTC) is a dynamical phase of matter that can be seen in the driven dynamics of spin models [23–27, 31].

A model where MBL-DTC is seen is the Floquet dynamics of a one-dimensional Ising model with Floquet operator

$$U_{\text{drive}}(\theta_x) = \left[\prod_{\ell=0}^{N-2} \exp \left(-i \varphi_{zz,\ell} \frac{1}{4} Z_\ell Z_{\ell+1} \right) \right] \left[\prod_{\ell=0}^{N-1} \exp \left(-i \varphi_{z,\ell} \frac{1}{2} Z_\ell \right) \right] \left[\prod_{\ell=0}^{N-1} \exp \left(-i \theta_x \frac{1}{2} X_\ell \right) \right], \quad (1)$$

where φ_z 's and φ_{zz} 's have spatial disorder and θ_x is the same for all spins and corresponds to a control parameter [27, 31]. This model shows MBL-DTC for $\theta_x > 0.84\pi$ and shows thermalization for $\theta_x < 0.84\pi$. I look for eigenstate ordering in the eigenvalues of $U_{\text{drive}}(\theta_x)$.

Since the MBL-DTC here corresponds to period doubling, the eigenstate order corresponds to the eigenvalues of $U_{\text{drive}}(\theta_x)$ differing by a phase of π : in the MBL-DTC dynamical phase, every eigenvalue η is paired with a partner $\eta e^{i\pi} = -\eta$ obtained by multiplying by the phase factor $e^{i\pi}$. This cat-doublet pairing is the emergent eigenstate-order signature of the period-doubled phase, exact in the ideal-pulse soluble limit and emergent though rigid for generic disorder, rather than a symmetry identity of the driven Hamiltonian [1–3].

I look for eigenstate order in a few instances of $U_{\text{drive}}(\theta_x)$.

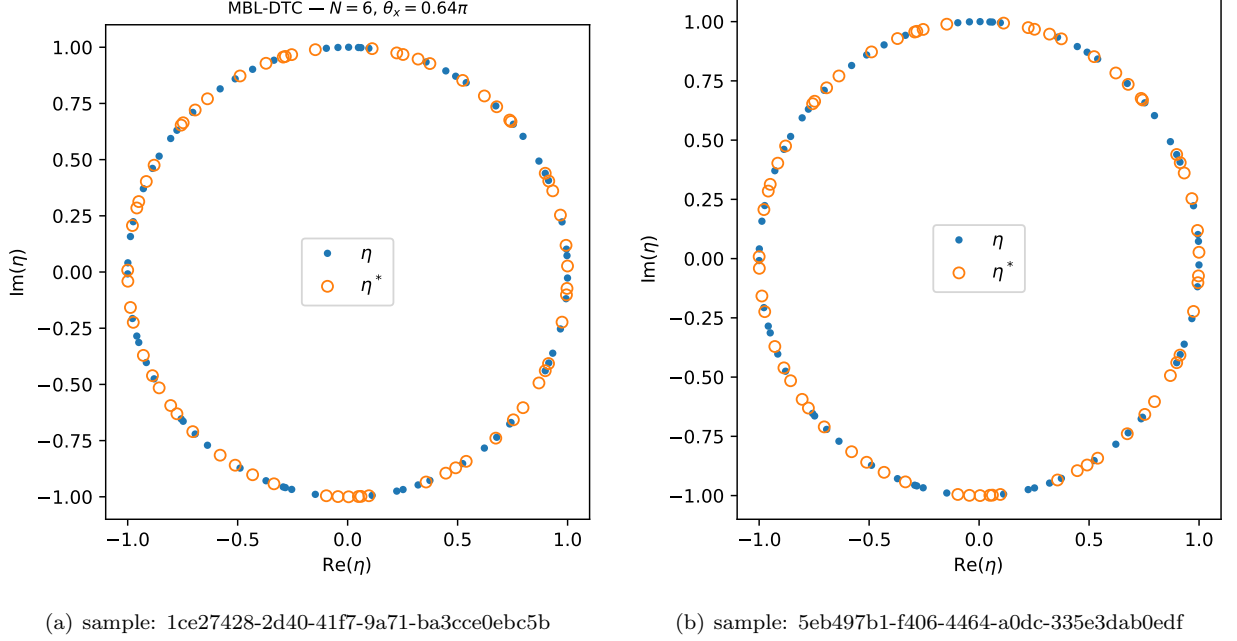


FIG. 1. $N = 6$, $\theta_x = 0.64\pi$. Eigenvalues η (filled) and their complex conjugates η^* (open), reflections about the real axis, of the Floquet operator; the phase- π partners $-\eta$ associated with period doubling are not shown.

IV. QUANTUM MANY-BODY SCARS

Quantum many-body scars (QMBS) were discovered as an explanation for periodic revivals of a Néel ordered initial state in a one-dimensional array of neutral atoms with ground-to-Rydberg excitation [5, 6].

The Hamiltonian of this system in the basis of $|g\rangle$, representing an atomic ground state and $|r\rangle$, representing an atomic Rydberg state is effectively an Ising model with transverse and longitudinal fields, which I call slanted field Ising (SFI) model. For simplicity I consider the case of interactions only between nearest neighbors when excited to atomic Rydberg states. In the rotating frame at the angular frequency of the laser driving the $|g\rangle \leftrightarrow |r\rangle$, transition and under the rotating wave approximation, the Hamiltonian reads

$$H_{\text{SFI}} = \frac{\Omega}{2} \sum_l X_l - \frac{\Delta}{2} \sum_l Z_l + V_{rr} \sum_l \left(\frac{1+Z_l}{2} \right) \left(\frac{1+Z_{l+1}}{2} \right), \quad (2)$$

where Ω is the Rabi frequency for, and Δ is the detuning from, the $|g\rangle \leftrightarrow |r\rangle$ transition, and V_{rr} is the interaction between nearest-neighbor atoms when excited to their Rydberg states.

A common regime of operation is the Rydberg blockade regime, which occurs when $|V_{rr}| \gg \Omega$. In the nearest neighbor Rydberg blockade and zero detuning $\Delta = 0$ regime, the system can be described using a PXP model [7, 8, 32], whose Hamiltonian reads

$$H_{\text{PXP}} = \Omega \sum_l P_{l-1} X_l P_{l+1}, \quad (3)$$

where P_l represents the projector onto the ground state, $|g\rangle\langle g|$ for the l -th atom, hence the name ‘PXP’ model. The PXP model is known to have QMBS. Therefore, I expect QMBS to be seen in the SFI model, in an appropriately constrained part of the Hilbert space.

To study this I consider quantum quenches, but instead of looking at the time evolution of an initial state, I consider the time evolution operators

$$U_{\text{PXP}}(t) = \exp(-itH_{\text{PXP}}), \quad (4)$$

and

$$U_{\text{SFI}}(t) = \exp(-itH_{\text{SFI}}), \quad (5)$$

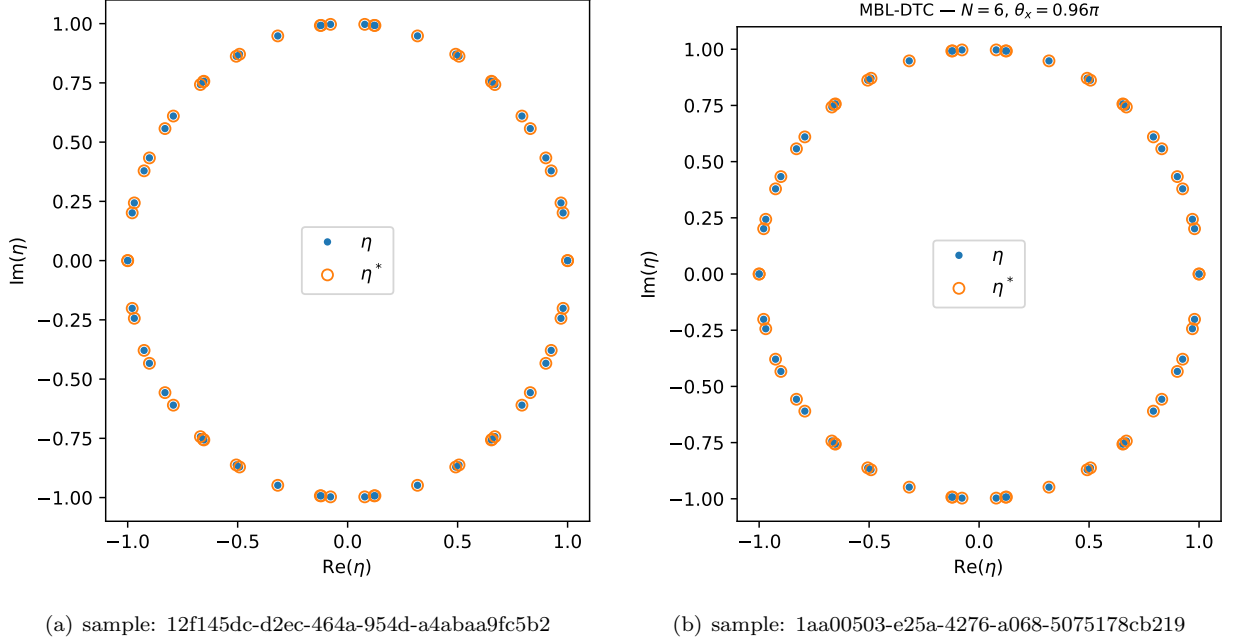


FIG. 2. $N = 6$, $\theta_x = 0.96\pi$. Eigenvalues η (filled) and their complex conjugates η^* (open), reflections about the real axis, of the Floquet operator; the phase- π partners $-\eta$ associated with period doubling are not shown.

for the PXP and SFI models respectively. I plot the eigenvalues, η of these time evolution operators and their complex conjugates η^* . The complex conjugate $\eta^* = e^{+iEt}$ is itself an eigenvalue of the propagator $U(t) = \exp(-itH)$ only when the spectrum of H contains $-E$ (up to phase aliasing of e^{-iEt}); it is guaranteed for every eigenvalue when the spectrum of H is symmetric under $E \leftrightarrow -E$, that is, when the model has an energy-reflection symmetry, that is, a unitary S with $SHS^\dagger = -H$ (equivalently, a chiral symmetry of the Floquet unitary); such a symmetry maps the eigenvalue $\eta = e^{-iEt}$ of the state at energy E to the eigenvalue e^{+iEt} of the partner state at energy $-E$.

I find that across different durations, t and system sizes N , the eigenvalues of $U_{\text{PXP}}(t)$ are complex conjugate pairs. This pairing is a symmetry consequence rather than an eigenstate-order signature: at $\Delta = 0$, the particle-hole symmetry operator $\Gamma = \prod_l Z_l$ anticommutes with every PXP term $P_{l-1}X_lP_{l+1}$, so $\{\Gamma, H_{\text{PXP}}\} = 0$ and the spectrum is symmetric under $E \leftrightarrow -E$ [7], which makes η^* the eigenvalue of the partner state at energy $-E$. The longitudinal detuning term $\Delta \sum_l Z_l/2$ of the general builder, Eq. (7), commutes with Γ and breaks this anticommutation; the PXP figures shown here are $\Delta = 0$ outputs of an earlier version of the package, with legacy filenames without an explicit detuning term (Sec. V E).

For $U_{\text{SFI}}(t)$, I find that a fraction of the eigenvalues occur close to their complex conjugates. The SFI Hamiltonian of Eq. (2) has no energy-reflection symmetry for $V_{rr} \neq 0$, so the conjugate points are plotted as reflections about the real axis, that is, as a guide to the eye, and no spectral mechanism guarantees an overlap. In the parameter regime of the figures ($\Delta = 0$, $V_{rr} = 100\Omega$) the absence of such a symmetry follows directly from the spectrum: the Rydberg-interaction term $V_{rr} \sum_l P_r^{(l)} P_r^{(l+1)}$ is positive semidefinite, so $E_{\min} \geq -\Omega N/2$, while the all-Rydberg product state has energy $V_{rr}(N-1)$, so $E_{\max} \geq V_{rr}(N-1) > \Omega N/2$; a spectrum invariant under $E \leftrightarrow -E$ would require $E_{\max} = -E_{\min} \leq \Omega N/2$, a contradiction. Any apparent overlap of η with η^* may reflect phase aliasing, that is, eigenphases of large-magnitude Et wrapping around the unit circle modulo 2π , rather than a spectral symmetry.

In summary, two distinct pairing mechanisms act in this work. A chiral, or particle-hole, symmetry, that is, a unitary $\mathcal{P}$ with $\mathcal{P}H\mathcal{P}^\dagger = -H$ (equivalently, $\{\mathcal{P}, H\} = 0$ for $\mathcal{P}^2 = \mathbb{I}$), forces the spectrum to be symmetric under $E \leftrightarrow -E$, so that propagator eigenvalues occur in conjugate pairs $\eta \leftrightarrow \eta^*$; this structure underlies the exact scarred eigenstates of PXP-like and fine-tuned models [7, 10, 11] and, for Floquet systems, follows from a chiral symmetry of the Floquet unitary [29]. The period-doubled MBL-DTC phase instead exhibits a cat-doublet structure that pairs quasienergies as $\varepsilon \leftrightarrow \varepsilon + \pi$, so that propagator eigenvalues occur in phase- π pairs $\eta \leftrightarrow -\eta$; this pairing is the emergent eigenstate-order signature of the phase rather than a symmetry identity [1, 2]. For a generic real-symmetric Hamiltonian, neither pairing is guaranteed.

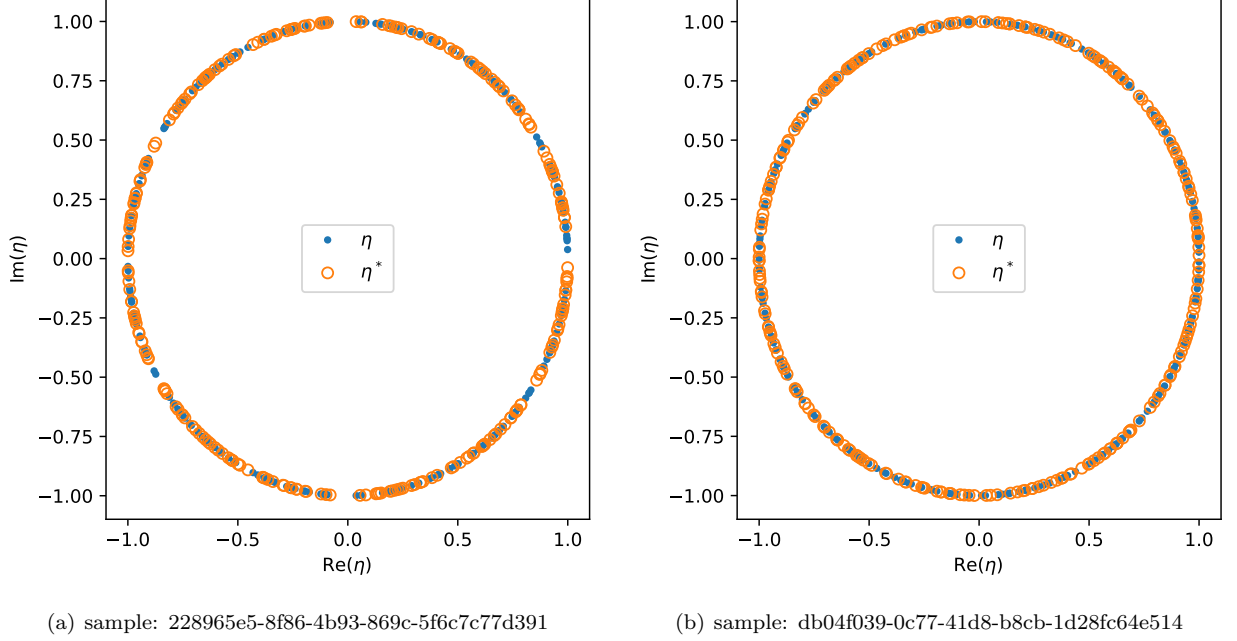


FIG. 3. $N = 8$, $\theta_x = 0.64\pi$. Eigenvalues η (filled) and their complex conjugates η^* (open), reflections about the real axis, of the Floquet operator; the phase- π partners $-\eta$ associated with period doubling are not shown.

V. NUMERICAL METHODS

All numerical results in this work are produced with the `mbl_eigen` package accompanying this manuscript. The package combines root-level command-line entry points with a reusable Python library. QuTiP provides the operator algebra, tensor products, operator expansion, and partial traces; NumPy and SciPy provide dense linear algebra and disorder sampling; PyTorch and JAX provide optional accelerator backends for Hermitian eigendecompositions; and an optional layer built on Qiskit constructs and simulates the same propagators as gate-native quantum circuits. The root-level `main*.py` scripts are compatibility shims: each builds its argument parser from the shared builders of `mbl_eigen.cli` and forwards the parsed arguments to a package-level runner.

A. Eigenvalue-analysis workflows

1. QMBS and PXP eigenphase analysis

The QMBS workflow (`mbl_eigen.qmbs_app.run_qmbs`) constructs two Hamiltonians for a chain of N spin-1/2 sites through the builders of `mbl_eigen.qmbs_model`: a slanted-field-Ising (SFI) Hamiltonian with a transverse drive, a longitudinal detuning term, and nearest-neighbor Rydberg-projector interactions, and a constrained PXP Hamiltonian in which each transverse drive term is flanked by ground-state projectors on the neighboring sites, with open boundary conditions. In the conventions of the implementation,

$$H_{\text{SFI}} = \frac{\Omega}{2} \sum_l X_l + \frac{\Delta}{2} \sum_l Z_l + V_{rr} \sum_l P_r^{(l)} P_r^{(l+1)}, \quad P_r = \frac{\mathbb{I} - Z}{2}, \quad (6)$$

and

$$H_{\text{PXP}} = \frac{\Omega}{2} \sum_l P_g^{(l-1)} X_l P_g^{(l+1)} + \frac{\Delta}{2} \sum_l Z_l, \quad P_g = \frac{\mathbb{I} + Z}{2}. \quad (7)$$

Both Hamiltonians are diagonalized with the selected Hermitian backend of the eigensolver layer described in Sec. V B. The energy eigenvalues E_n are converted into unitary eigenvalues $\eta_n = \exp(-iE_n t)$ for the requested duration t ; the

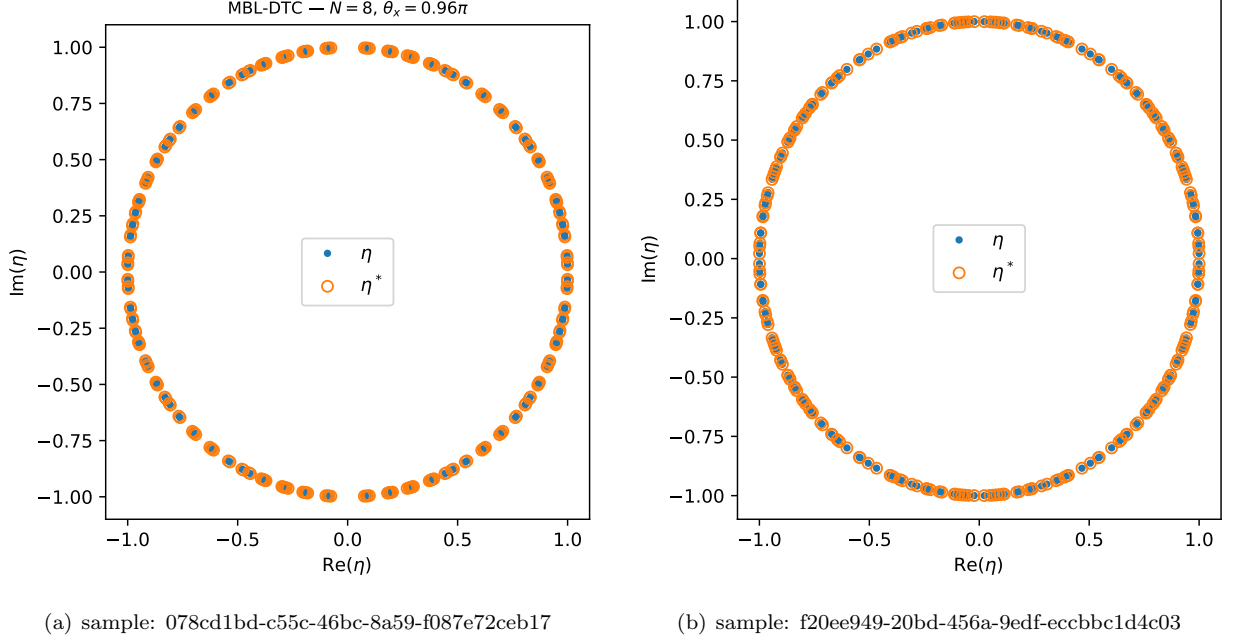


FIG. 4. $N = 8$, $\theta_x = 0.96\pi$. Eigenvalues η (filled) and their complex conjugates η^* (open), reflections about the real axis, of the Floquet operator; the phase- π partners $-\eta$ associated with period doubling are not shown.

eigenphases $\arg \eta_n$ are mapped to $[0, 2\pi)$ and sorted; and the mean adjacent level-spacing ratio defined in Sec. V C is computed both for the energy spectra and for the eigenphase spectra. Finally, the workflow plots the eigenvalues η_n together with their complex conjugates η_n^* on the unit circle and writes one figure per Hamiltonian. The shipped parameter values are $\Omega = 1$ and $V_{rr} = 100 \Omega$.

2. MBL-DTC Floquet analysis

The MBL-DTC workflow (`mb1_eigen.mbl_dtc_app.run_mbl_dtc`) samples the on-site rotation angles $\varphi_{z,\ell}$ and the nearest-neighbor interaction angles $\varphi_{zz,\ell}$ independently and uniformly from $[0, \pi)$, and assembles the Floquet operator of one drive period as an ordered product of single-site rotation exponentials and nearest-neighbor interaction exponentials,

$$U_{\text{drive}} = \left[\prod_{\ell=0}^{N-2} \exp \left(-i \frac{\varphi_{zz,\ell}}{2} Z_\ell Z_{\ell+1} \right) \right] \left[\prod_{\ell=0}^{N-1} \exp \left(-i \frac{\varphi_{z,\ell}}{2} Z_\ell \right) \right] \left[\prod_{\ell=0}^{N-1} \exp \left(-i \frac{\theta_x}{2} X_\ell \right) \right], \quad (8)$$

where each single-site factor is promoted to the full Hilbert space with `qutip.qip.operations.expand_operator`. Because the Floquet operator is unitary rather than Hermitian, it is diagonalized with the general eigenproblem solver of Sec. V B, which currently supports the QuTiP backend only. The eigenphases are extracted and sorted as in the QMBS workflow, the level-spacing ratio is computed from the Floquet eigenphases, the eigenphases of the squared Floquet operator U_{drive}^2 are written to the logging output, and the eigenvalues η are plotted together with their complex conjugates η^* , that is, their reflections across the real axis, on the unit circle. The phase- π partners $-\eta$ associated with period doubling are a related structure and are not plotted separately; the two descriptions coincide at the DTC point of the dipolar experiment of Ref. [23], where the two-to-one Floquet quasienergies pair as $\pm \varepsilon_i$ and cluster near $\varepsilon_i \approx \pi/2$, so that the complex conjugate η^* and the phase- π partner $-\eta$ of an eigenvalue are numerically equal.

3. Random-field MBL workflows

Three workflows study a random-field MBL model built by `mb1_eigen.mbl_model`. One disorder realization consists of nearest-neighbor couplings J_i drawn from a normal distribution with mean J and standard deviation σ_J for

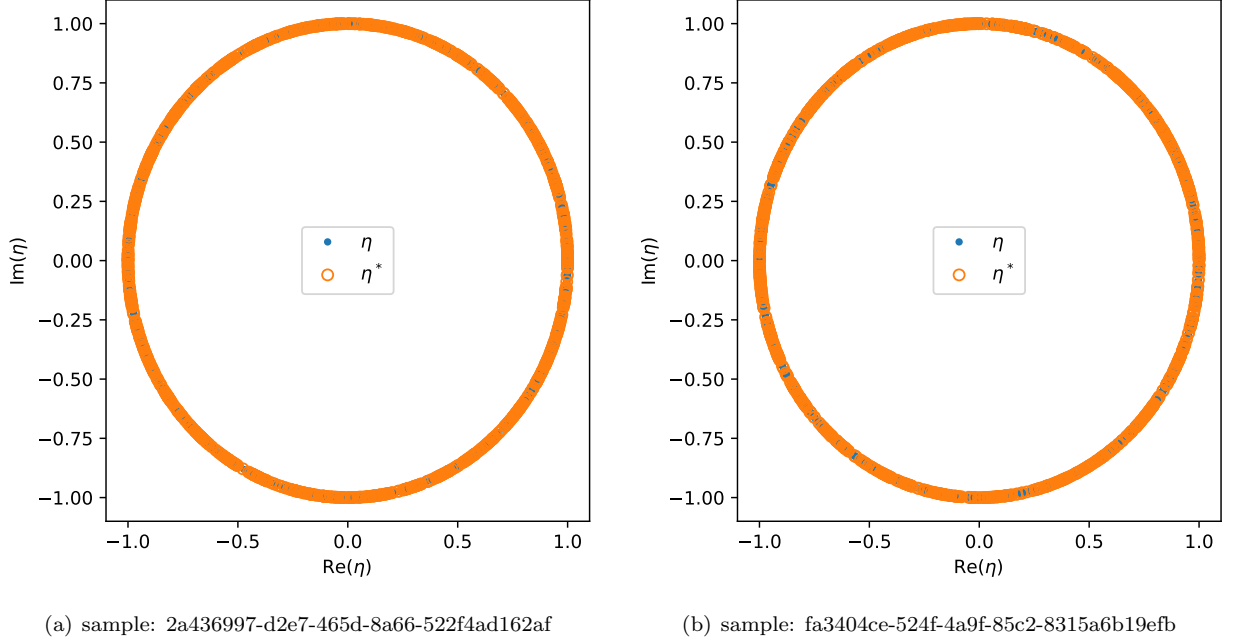


FIG. 5. $N = 10$, $\theta_x = 0.64\pi$. Eigenvalues η (filled) and their complex conjugates η^* (open), reflections about the real axis, of the Floquet operator; the phase- π partners $-\eta$ associated with period doubling are not shown.

$i = 0, \dots, N - 2$, local field magnitudes B_i drawn from a normal distribution with mean B and standard deviation σ_B , and field polar angles θ_i drawn uniformly from $[\theta_{\min}\pi, \theta_{\max}\pi]$. The Hamiltonian reads

$$H_{\text{MBL}} = \sum_i B_i \sin \theta_i X_i + \sum_i B_i \cos \theta_i Z_i + \sum_{i=0}^{N-2} J_i Z_i Z_{i+1}. \quad (9)$$

The spectrum workflow (`mbl_eigen.mbl_app.run_mbl`) diagonalizes H_{MBL} , computes the half-chain von Neumann entanglement entropy of every eigenstate by partial trace over the first $\lfloor N/2 \rfloor$ sites (`qutip.ptrace` and `qutip.entropy_vn`), computes the level-spacing ratios of the energy spectrum and of the derived eigenphases, and plots the entanglement entropy against energy together with the maximally mixed value $\lfloor N/2 \rfloor \ln 2$ as a reference line.

The dynamics workflow (`mbl_eigen.mbl_app.run_mbl_dynamics`) expands the initial product state $|1 \dots 1\rangle$ in the Hamiltonian eigenbasis, evaluates the return amplitude

$$A(t) = \sum_n |\langle E_n | \psi_0 \rangle|^2 e^{-iE_n t} \quad (10)$$

on a uniform time grid with nominal spacing 0.0625 up to the requested duration, and plots the return rate $\lambda(t) = -\ln |A(t)|^2 / N$.

The propagator workflow (`mbl_eigen.mbl_app.run_mbl_propagator`) converts the Hamiltonian to a dense array, expands the local operators X_i , Y_i , and Z_i to every site normalized by $1/\sqrt{2^N}$, rotates them into the eigenbasis through $V^\dagger O V$, where V is the eigenvector matrix, evolves each rotated operator as $\tilde{O}(t) = V \exp(-iDt) \tilde{O} \exp(iDt) V^\dagger$, with D the diagonal matrix of phases $\exp(-iE_n t)$, so that $O(t) = U(t) O U(t)^\dagger$ with $U(t) = V \exp(-iDt) V^\dagger$, and computes the overlap matrix $\text{tr} \left[O_\alpha^{(i)} O_\beta^{(j)}(t) \right]$ between the initial and time-evolved local operators over all x , y , and z channels, sites, and time points on the same grid. The final-time overlap matrix is written to the logging output; no figure is saved.

A remark on conventions is in order. The normalization conventions of the builders described in this section differ from the model definitions of Secs. III and IV: the interaction exponential in Eq. (8) carries twice the rotation angle of the corresponding factor in Eq. (1); the detuning term in Eq. (6) enters with the opposite sign, and the Rydberg projector is $(\mathbb{I} - Z)/2$ rather than $(\mathbb{I} + Z)/2$ as in Eq. (2); and the PXP builder of Eq. (7) contains a longitudinal detuning term and a factor of one half that are absent from Eq. (3). The builders described in this section are the definitive reference for all numerical results presented in this manuscript.

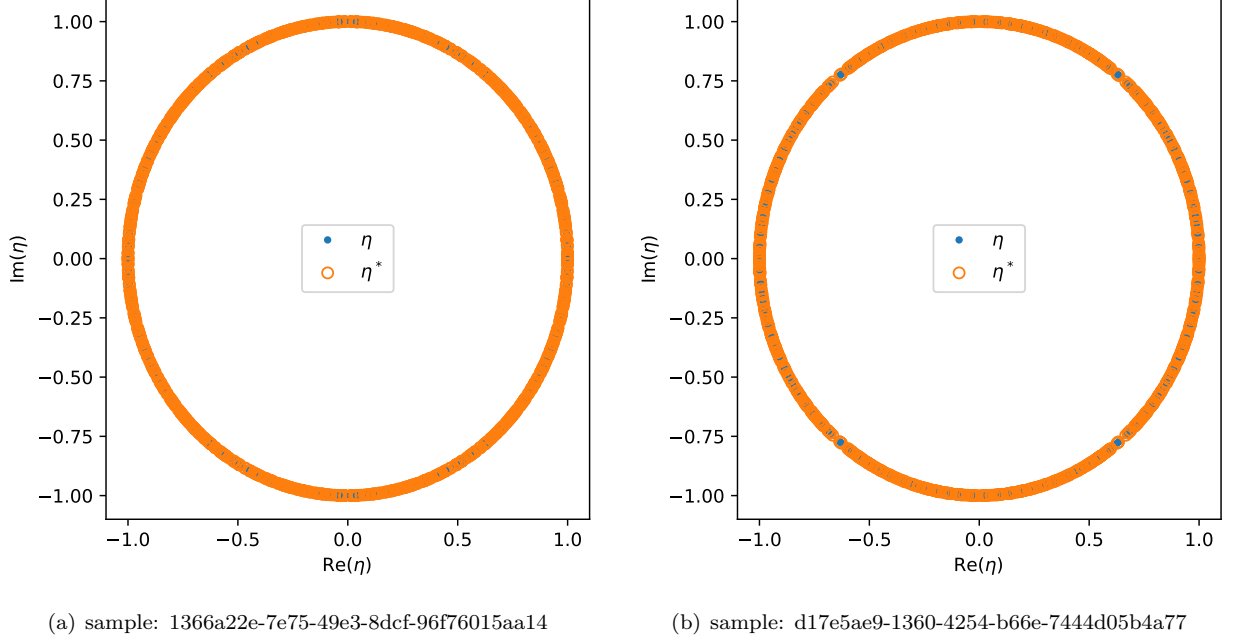


FIG. 6. $N = 10$, $\theta_x = 0.96\pi$. Eigenvalues η (filled) and their complex conjugates η^* (open), reflections about the real axis, of the Floquet operator; the phase- π partners $-\eta$ associated with period doubling are not shown.

B. Eigensolver backends

The eigensolver layer (`mb1_eigen.eigensolver`) exposes two entry points: `solve_hermitian_eigenproblem`, which supports the backends `qobj` (QuTiP), `numpy`, `scipy`, `torch`, and `jax`; and `solve_general_eigenproblem` for non-Hermitian operators, which currently supports the QuTiP backend only. Before diagonalization the operator is converted to a complex128 dense array and tested for Hermiticity; if the test fails, the operator is symmetrized as $(H + H^\dagger)/2$ and a warning is logged. Hermitian results are converted with `numpy.real_if_close`, eigenvalues with non-negligible imaginary parts are rejected rather than silently discarded, and the eigenvalues are returned in ascending order with the eigenvector columns reordered consistently. Requests without eigenvectors use each backend's eigenvalue-only routine. The returned `EigenResult` carries the backend and device labels and provides helpers to rebuild QuTiP kets and basis-change operators from the raw eigenvector matrix.

Devices are selected with the choices `auto`, `cpu`, `gpu`, `cuda`, and `mps`. The QuTiP, NumPy, and SciPy backends run on the CPU only; the PyTorch backend resolves CUDA devices and rejects the Metal performance-shaders path for eigendecomposition; and the JAX backend enables float64 arithmetic and maps explicit `cuda` and `mps` requests to a matching accelerator platform. PyTorch and JAX are optional installation extras and are imported lazily, only when their backends are selected.

C. Level-spacing statistics

The module `mb1_eigen.level_repulsion` computes the mean adjacent level-spacing ratio. Given a sorted spectrum $\{\lambda_n\}$, the routine removes a fraction `fraction_cutoff` of the levels from each edge of the spectrum (0.02 by default; the eigenvalue-analysis workflows use zero, that is, the full spectrum), forms the adjacent spacings $s_n = \lambda_{n+1} - \lambda_n$ over the remaining bulk, and returns

$$r = \left\langle \frac{\min(s_n, s_{n+1})}{\max(s_n, s_{n+1})} \right\rangle_n, \quad (11)$$

where the average runs over the bulk spacings. For eigenphase spectra the optional `circular_period` argument, set to 2π by the workflows, appends the wraparound spacing $\lambda_0 + 2\pi - \lambda_{\max}$ so that the statistic is well defined for phases on the circle. Spectra with fewer than three bulk levels, or with a zero adjacent spacing, raise an error; a lenient

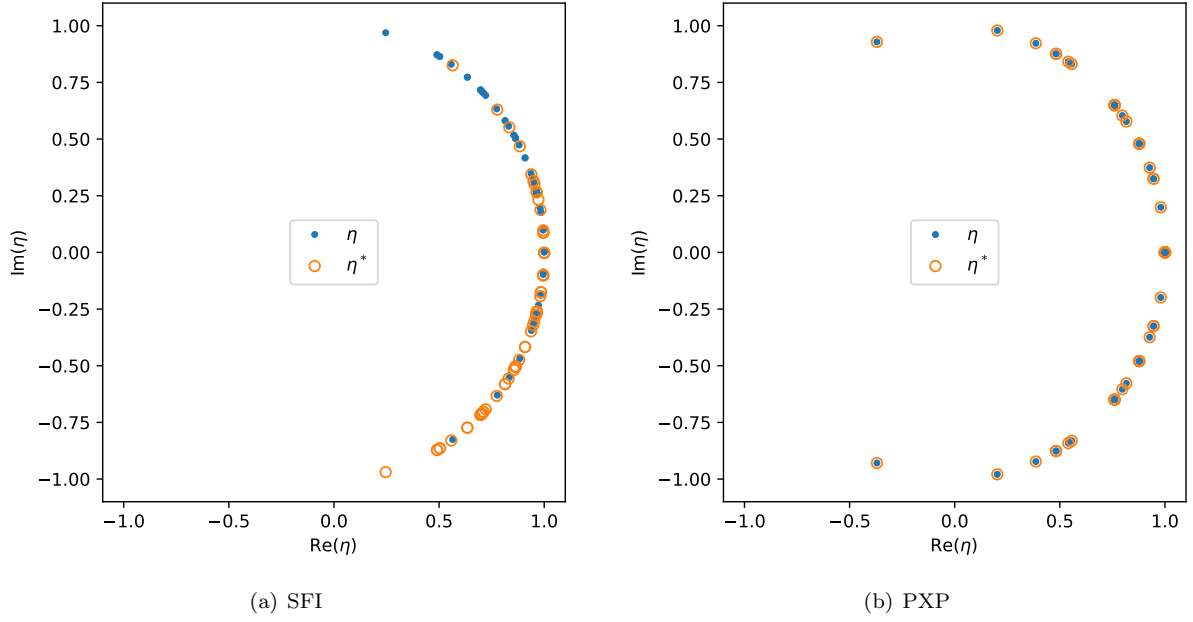


FIG. 7. Eigenvalues η (filled) and their complex conjugates η^* (open) of the propagator, plotted in the complex plane for $N = 6$, $\Omega t = 0.5$. (a) SFI with $V_{rr} = 100\Omega$: the conjugates are reflections about the real axis, shown as a guide to the eye. (b) PXP: the particle-hole symmetry $\Gamma = \prod_l Z_l$ of the PXP model at $\Delta = 0$ makes η^* the eigenvalue of the partner state at energy $-E$.

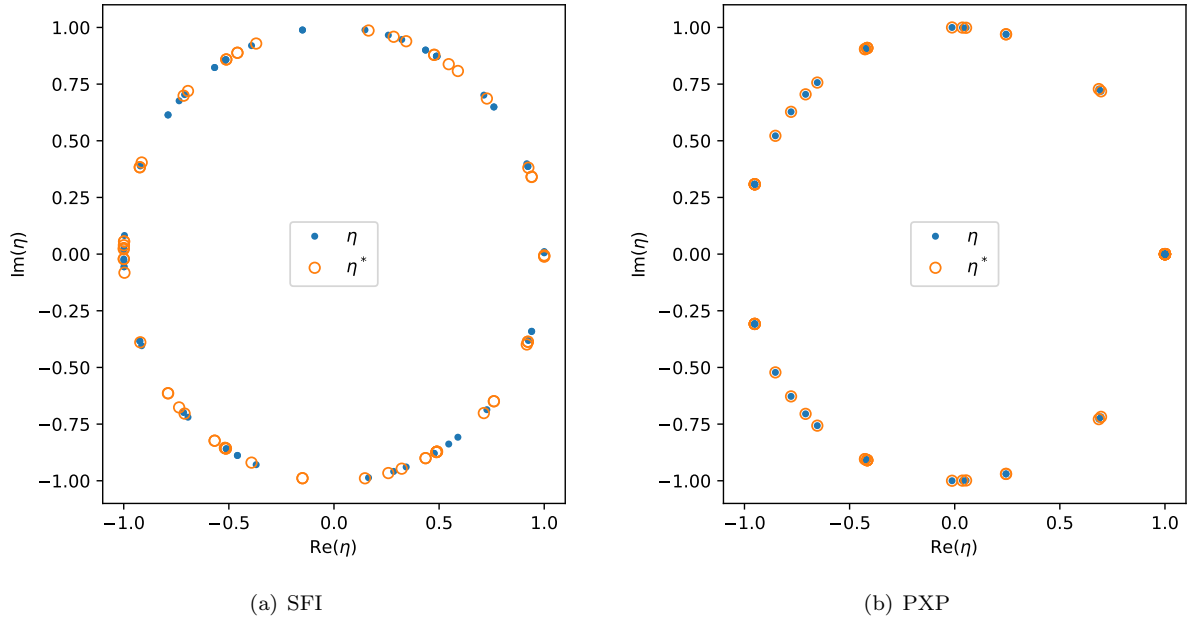


FIG. 8. Eigenvalues η (filled) and their complex conjugates η^* (open) of the propagator, plotted in the complex plane for $N = 6$, $\Omega t = 2$. (a) SFI with $V_{rr} = 100\Omega$: the conjugates are reflections about the real axis, shown as a guide to the eye. (b) PXP: the particle-hole symmetry $\Gamma = \prod_l Z_l$ of the PXP model at $\Delta = 0$ makes η^* the eigenvalue of the partner state at energy $-E$.

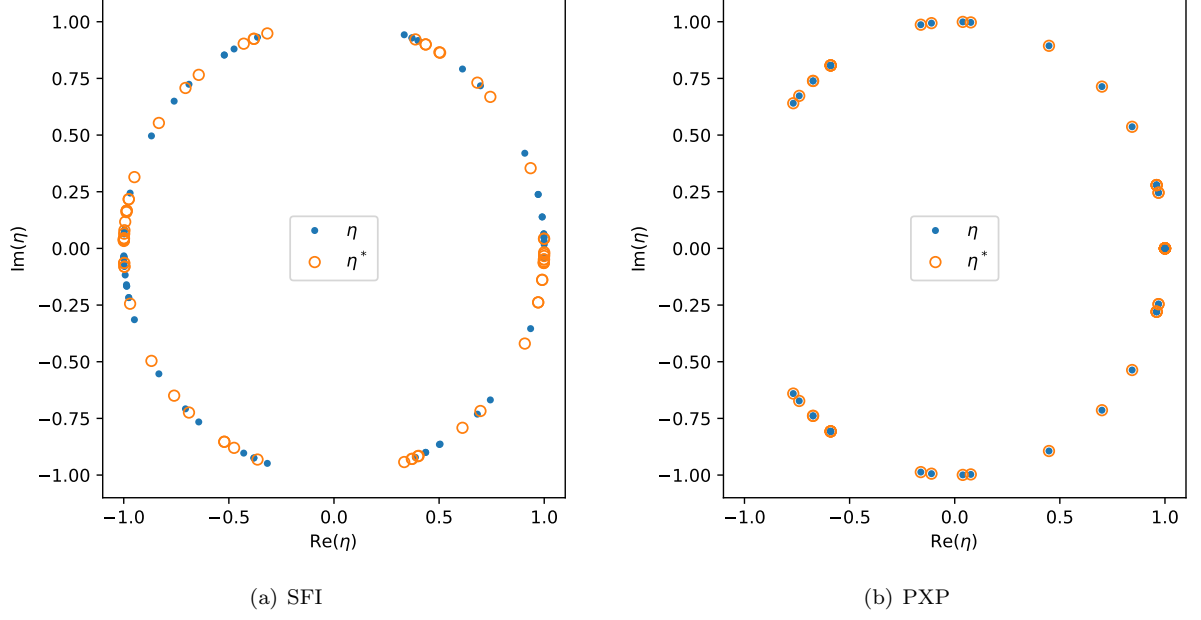


FIG. 9. Eigenvalues η (filled) and their complex conjugates η^* (open) of the propagator, plotted in the complex plane for $N = 6$, $\Omega t = 6$. (a) SFI with $V_{rr} = 100\Omega$: the conjugates are reflections about the real axis, shown as a guide to the eye. (b) PXP: the particle-hole symmetry $\Gamma = \prod_l Z_l$ of the PXP model at $\Delta = 0$ makes η^* the eigenvalue of the partner state at energy $-E$.

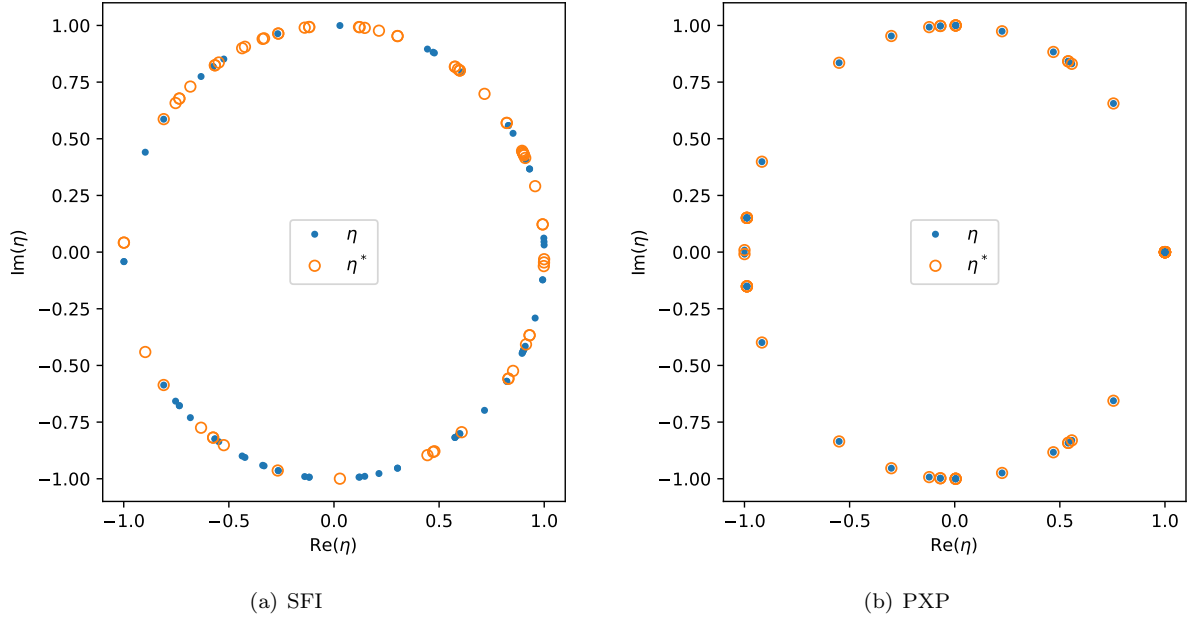


FIG. 10. Eigenvalues η (filled) and their complex conjugates η^* (open) of the propagator, plotted in the complex plane for $N = 6$, $\Omega t = 11$. (a) SFI with $V_{rr} = 100\Omega$: the conjugates are reflections about the real axis, shown as a guide to the eye. (b) PXP: the particle-hole symmetry $\Gamma = \prod_l Z_l$ of the PXP model at $\Delta = 0$ makes η^* the eigenvalue of the partner state at energy $-E$.

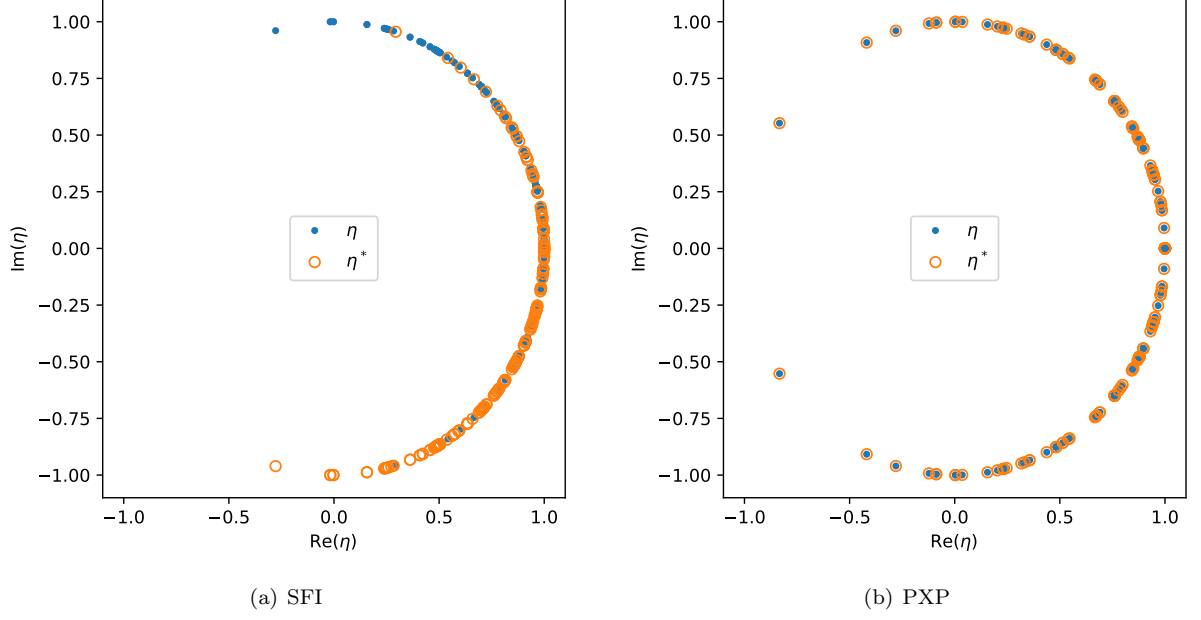


FIG. 11. Eigenvalues η (filled) and their complex conjugates η^* (open) of the propagator, plotted in the complex plane for $N = 8$, $\Omega t = 0.5$. (a) SFI with $V_{rr} = 100\Omega$: the conjugates are reflections about the real axis, shown as a guide to the eye. (b) PXP: the particle-hole symmetry $\Gamma = \prod_l Z_l$ of the PXP model at $\Delta = 0$ makes η^* the eigenvalue of the partner state at energy $-E$.

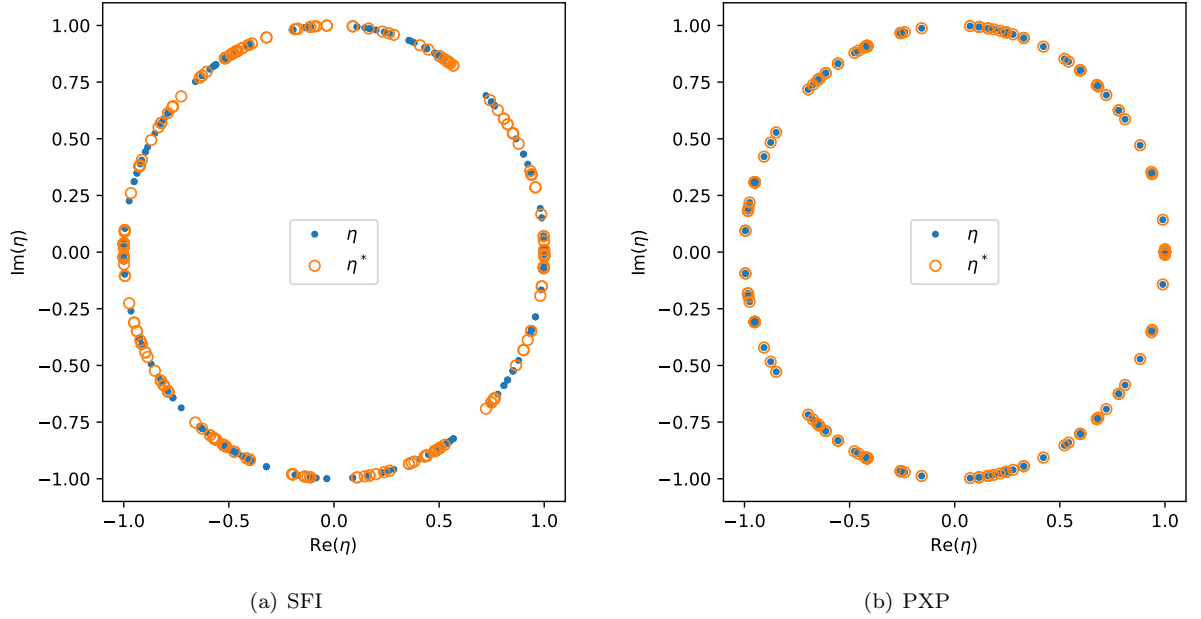


FIG. 12. Eigenvalues η (filled) and their complex conjugates η^* (open) of the propagator, plotted in the complex plane for $N = 8$, $\Omega t = 2$. (a) SFI with $V_{rr} = 100\Omega$: the conjugates are reflections about the real axis, shown as a guide to the eye. (b) PXP: the particle-hole symmetry $\Gamma = \prod_l Z_l$ of the PXP model at $\Delta = 0$ makes η^* the eigenvalue of the partner state at energy $-E$.

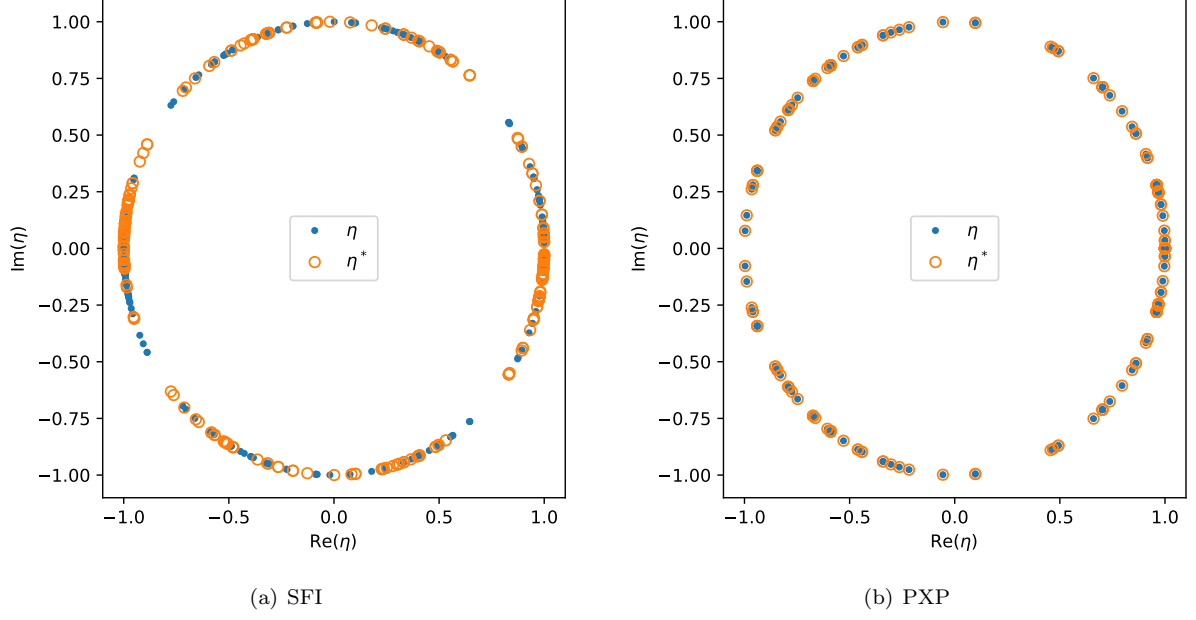


FIG. 13. Eigenvalues η (filled) and their complex conjugates η^* (open) of the propagator, plotted in the complex plane for $N = 8$, $\Omega t = 6$. (a) SFI with $V_{rr} = 100\Omega$: the conjugates are reflections about the real axis, shown as a guide to the eye. (b) PXP: the particle-hole symmetry $\Gamma = \prod_l Z_l$ of the PXP model at $\Delta = 0$ makes η^* the eigenvalue of the partner state at energy $-E$.

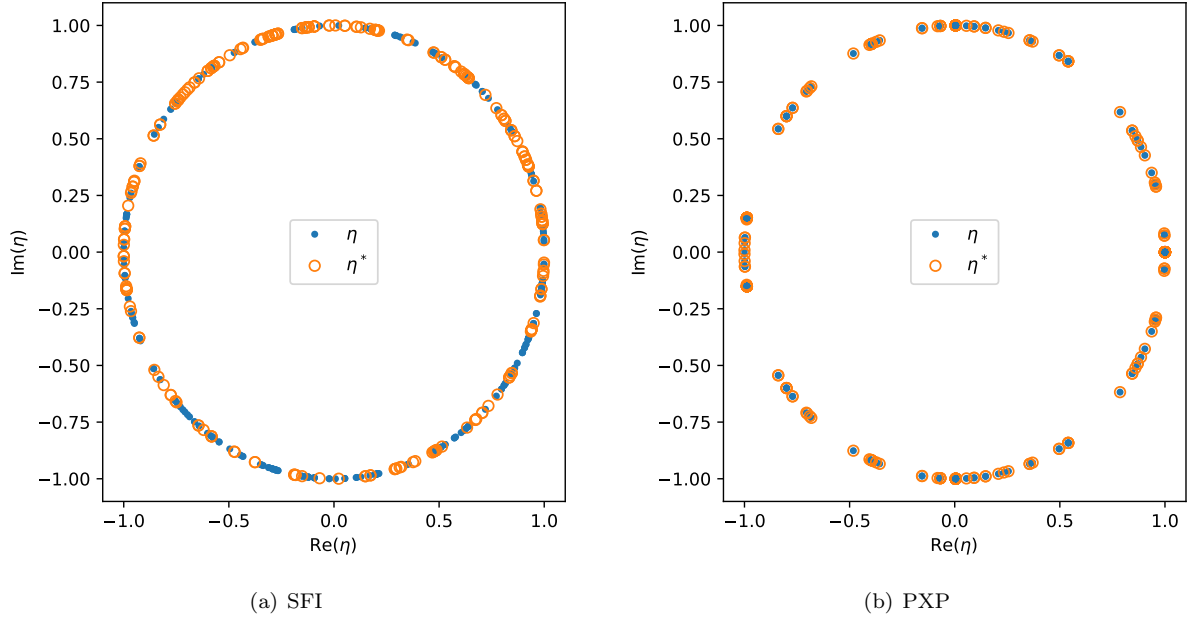


FIG. 14. Eigenvalues η (filled) and their complex conjugates η^* (open) of the propagator, plotted in the complex plane for $N = 8$, $\Omega t = 11$. (a) SFI with $V_{rr} = 100\Omega$: the conjugates are reflections about the real axis, shown as a guide to the eye. (b) PXP: the particle-hole symmetry $\Gamma = \prod_l Z_l$ of the PXP model at $\Delta = 0$ makes η^* the eigenvalue of the partner state at energy $-E$.

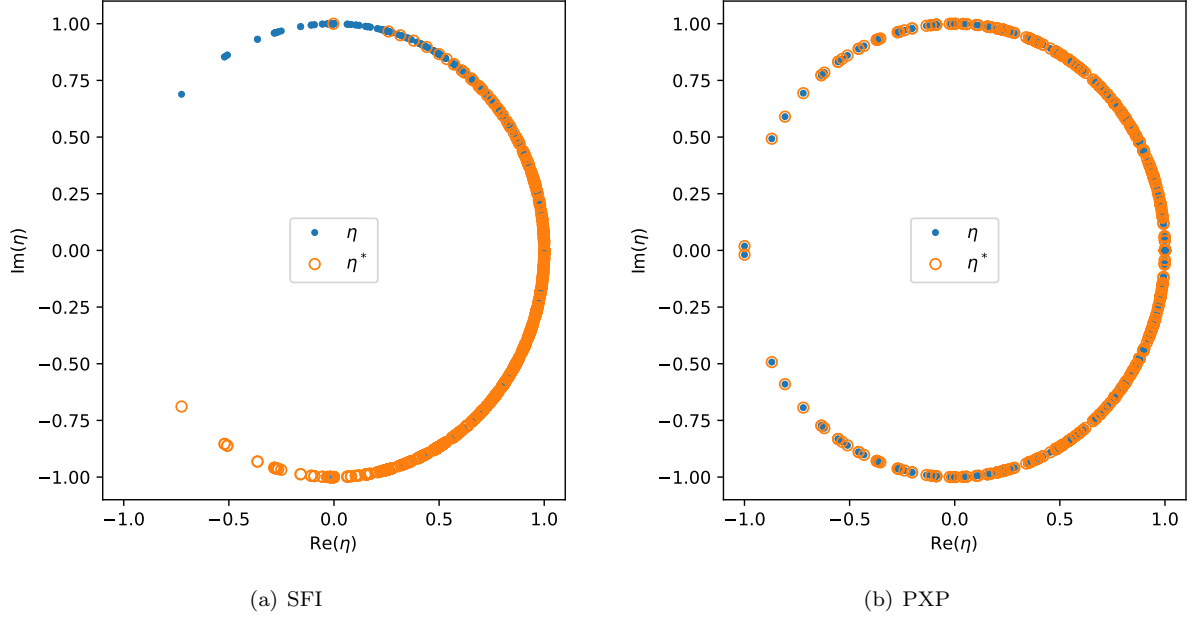


FIG. 15. Eigenvalues η (filled) and their complex conjugates η^* (open) of the propagator, plotted in the complex plane for $N = 10$, $\Omega t = 0.5$. (a) SFI with $V_{rr} = 100\Omega$: the conjugates are reflections about the real axis, shown as a guide to the eye. (b) PXP: the particle-hole symmetry $\Gamma = \prod_l Z_l$ of the PXP model at $\Delta = 0$ makes η^* the eigenvalue of the partner state at energy $-E$.

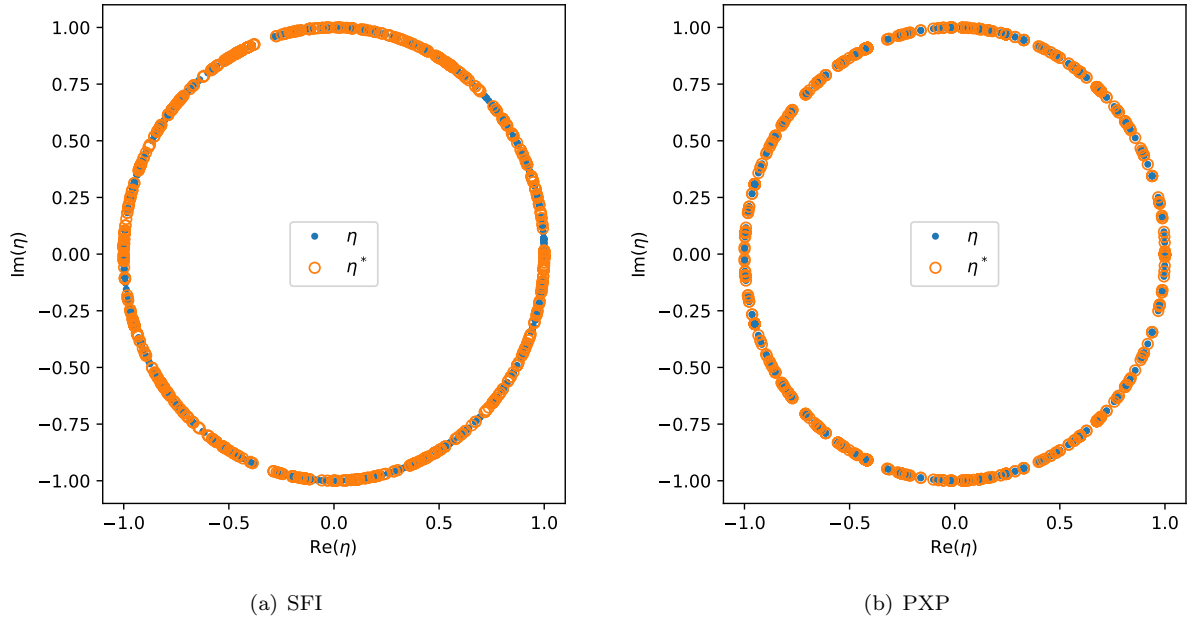


FIG. 16. Eigenvalues η (filled) and their complex conjugates η^* (open) of the propagator, plotted in the complex plane for $N = 10$, $\Omega t = 2$. (a) SFI with $V_{rr} = 100\Omega$: the conjugates are reflections about the real axis, shown as a guide to the eye. (b) PXP: the particle-hole symmetry $\Gamma = \prod_l Z_l$ of the PXP model at $\Delta = 0$ makes η^* the eigenvalue of the partner state at energy $-E$.

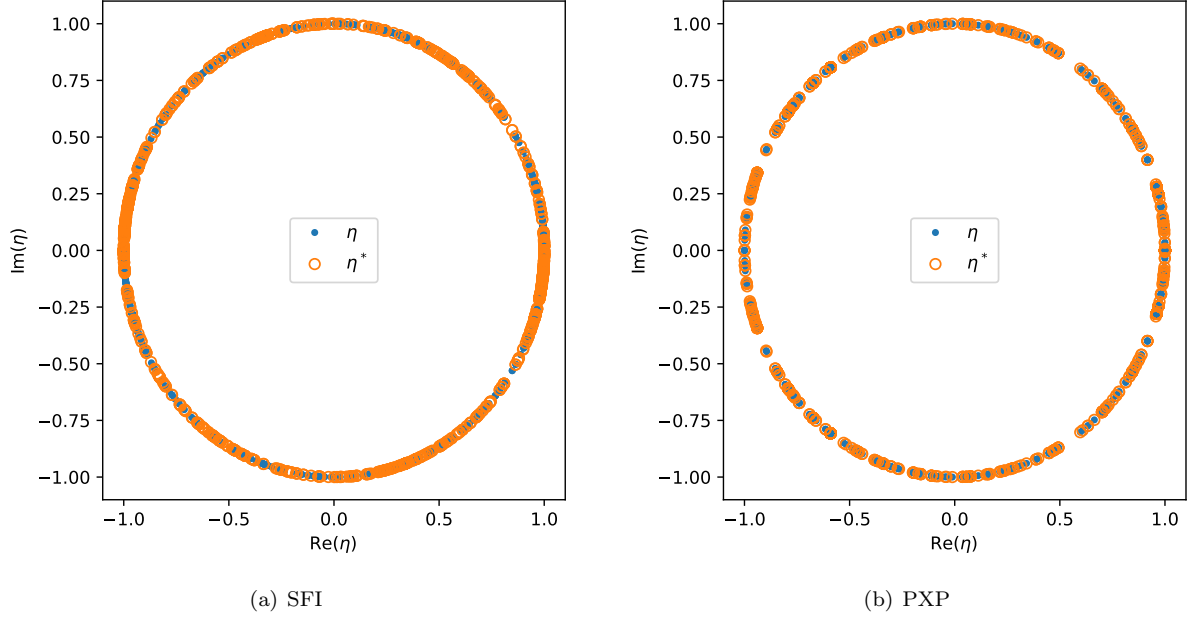


FIG. 17. Eigenvalues η (filled) and their complex conjugates η^* (open) of the propagator, plotted in the complex plane for $N = 10$, $\Omega t = 6$. (a) SFI with $V_{rr} = 100\Omega$: the conjugates are reflections about the real axis, shown as a guide to the eye. (b) PXP: the particle-hole symmetry $\Gamma = \prod_l Z_l$ of the PXP model at $\Delta = 0$ makes η^* the eigenvalue of the partner state at energy $-E$.

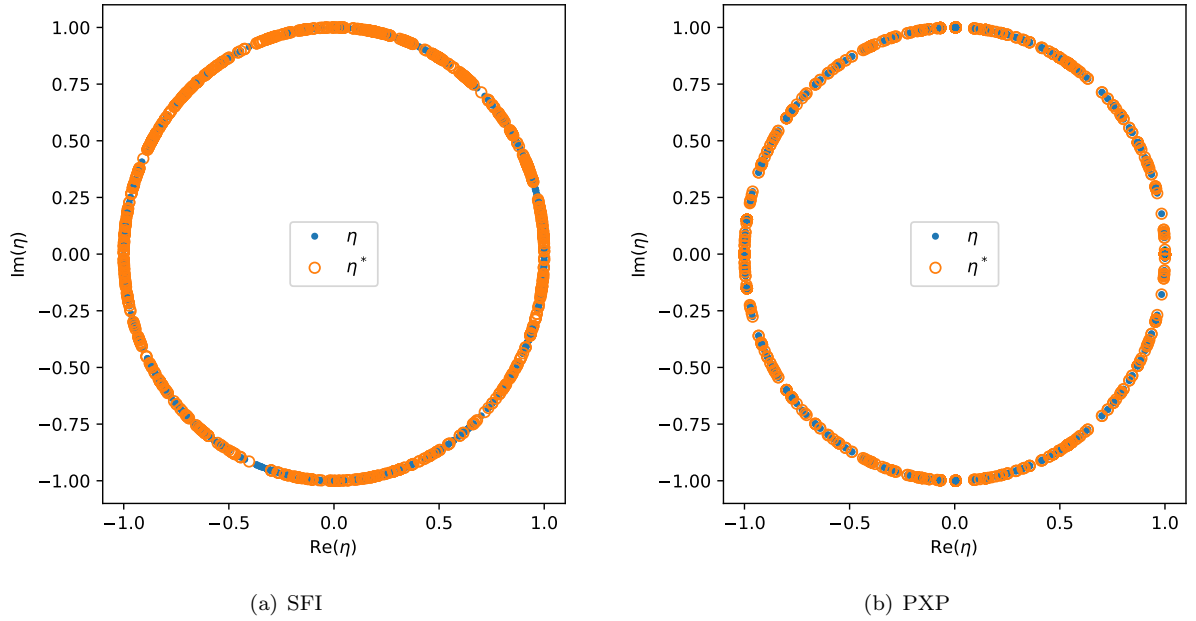


FIG. 18. Eigenvalues η (filled) and their complex conjugates η^* (open) of the propagator, plotted in the complex plane for $N = 10$, $\Omega t = 11$. (a) SFI with $V_{rr} = 100\Omega$: the conjugates are reflections about the real axis, shown as a guide to the eye. (b) PXP: the particle-hole symmetry $\Gamma = \prod_l Z_l$ of the PXP model at $\Delta = 0$ makes η^* the eigenvalue of the partner state at energy $-E$.

wrapper used by the QMBS and MBL-DTC workflows maps these degenerate cases to a not-a-number result with a logged warning, so that a run can continue past a degenerate model.

D. Qiskit circuit layer

An optional layer built on Qiskit (`mb1.eigen.qiskit.propagators` and `mb1.eigen.qiskit.simulation`) targets gate-based quantum computers directly. The MBL-DTC Floquet operator of Eq. (8) is already gate-native: the builder `build_mbl_dtc_floquet_circuit` implements it exactly as a circuit of `rx`, `rz`, and `rz` gates in the same order as the state-vector workflow, a global X rotation layer, then on-site Z rotations, then nearest-neighbor ZZ interactions, repeated for a configurable number of drive cycles, and requires no Trotterization. Time evolution under the random-field MBL Hamiltonian of Eq. (9) is synthesized as a nearest-neighbor Trotter circuit (`build_mbl_trotter_step_circuit`, `build_mbl_trotter_circuit`) in either first-order form, a diagonal layer followed by a transverse layer, or second-order form, a half diagonal layer, a transverse layer, and a half diagonal layer, with `rz` and `rz` gates implementing the diagonal Z/ZZ sector and `rx` gates the transverse sector. Model site ℓ is mapped to Qiskit qubit $N - 1 - \ell$, so that the circuit matrices follow the same site ordering as the QuTiP tensor operators.

The simulation engine supports three backends: an exact statevector backend that requires only the base Qiskit package; an `AerSimulator` backend that runs either in exact statevector mode or shot-based, requiring the optional `qiskit-aer` extra; and a fake-backend mode that attaches the noise model of an IBM fake provider, by default `FakeManilaV2`, and is always shot-based. For each requested time point the number of Trotter steps scales as $\max(1, \text{round}(n_{\text{trotter}}|t|))$. The engine returns the time grid, the return rate, that is, the Loschmidt echo $|\langle\psi_0|\psi(t)\rangle|^2$ in the all-ones initial state, and the site-resolved longitudinal magnetization $\langle Z_i \rangle(t)$; shot-based observables are estimated from the empirical count distributions.

E. Output conventions and reproducibility

Output filenames are generated centrally in `mb1.eigen.output_names`. Every workflow appends a random UUID suffix to each filename, so repeated runs intentionally produce different filenames. The MBL-DTC and MBL workflows accept a `--seed` flag that makes the sampled disorder reproducible. Regression tests for the package live in `tests/` and run with the standard `unittest` discovery mechanism. Figures produced by earlier versions of the package use legacy filenames without an explicit detuning term; the QMBS figures in this manuscript reference such legacy outputs.

VI. RESULTS: ENTANGLEMENT ENTROPY AND LEVEL-SPACING STATISTICS

TODO

VII. FINITE-SIZE SCALING AND THERMODYNAMIC LIMIT

TODO

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